

# Borg (Bring yer Own GRaphics): An Open-Source Tile-Based GPU with Silicon Tapeout

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## Motivation

- Every commercial GPU hides its microarchitecture behind NDAs
- This blocks **reproducibility**, **teaching**, and architectural experiments
- Concise:** Hardware (**1513 LoC** Scala), Compiler (**1600 LoC** Python), Firmware (**2000 LoC** C). Total stack **< 6k lines**.
- Open GPU simulators exist — but *no open GPU has reached silicon*

**Borg** is the fully open answer: a TBDR GPU SoC in Chisel, built with an open toolchain, and **taped out twice on real silicon**. Borg has contributed **68 patches upstream** to the open EDA ecosystem. Integrating the **TinyQV** RISC-V core (by Michael Bell), Borg provides a complete system-level open-source hardware reference.

To our knowledge: the only open-source GPU with a completed ASIC tapeout.

## Development Roadmap: Phases 1–6

- Phase 1** ✓ — **Foundational Architecture** (2025)
- Phase 2** [wip] — **GPU Autonomy / 28-Day Sprint**
- Phase 3** — **Linux-Capable CPU** (M/A/MMU)
- Phase 4** — **Mesa Vulkan Driver** (NIR → SPIR-B)
- Phase 5** — **Hardware Extensions** (SIMD, Blending)
- Phase 6** — **Vulkan 1.0 Conformance**

### Status Key Phase 2 Milestones

- ✓ 32-bit RISC-V ISA + Perspective Proj.
- ✓ Triangle Clipping + Fragment Interpolation
- ✓ Pixel Iterator (Rasterizer) + Z-Buffer
- ✓ Command FIFO + SystemRDL Register Map
- ✓ Texture Fetch (PSRAM) + Shared Mem. Ctrl
- [wip] GPU DMA Engine + VGA Output

## Design Values

**Transparency** No binary blobs, no hidden firmware.  
**Accessibility** Runs on a \$20 FPGA; uses \$500 ASIC runs.  
**Simplicity** Minimalist TBDR pipeline for maximum performance/area.

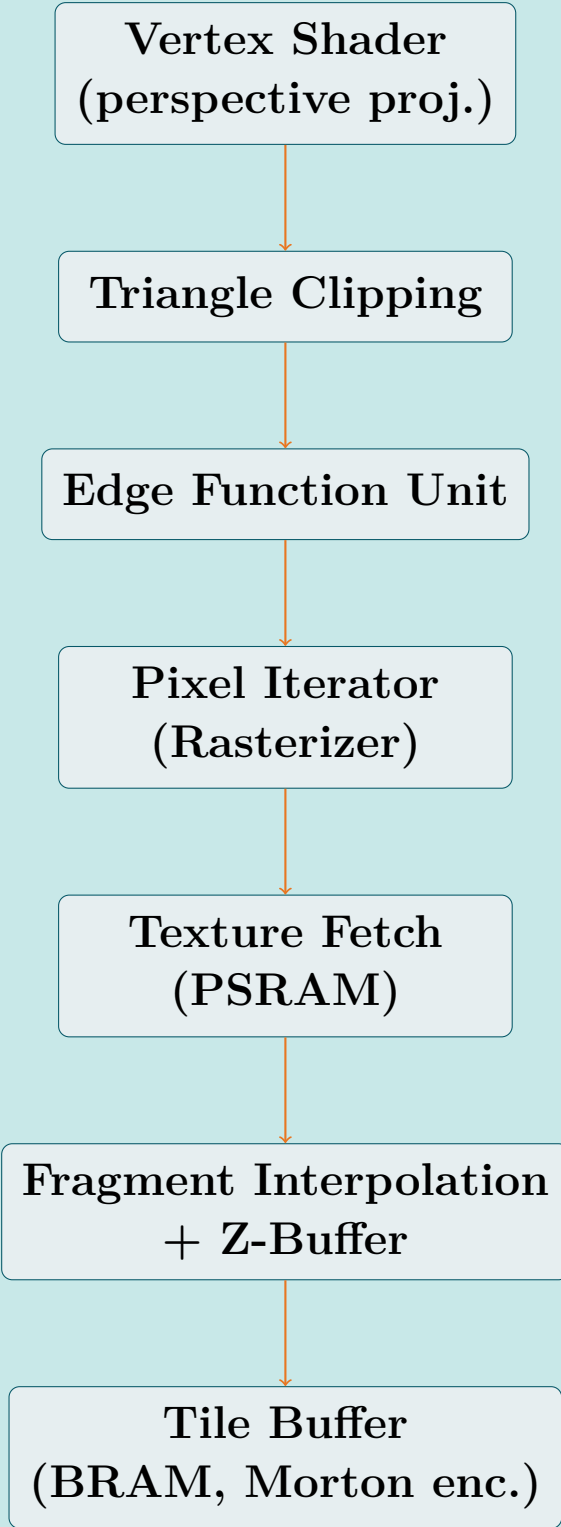
## The Transparency Gap: Borg vs. Commercial

| Feature        | Borg          | Mobile      | Desktop     |
|----------------|---------------|-------------|-------------|
| RTL Source     | <b>Open</b>   | Proprietary | Proprietary |
| EDA Toolchain  | <b>Libre</b>  | Proprietary | Proprietary |
| Silicon Layout | <b>Public</b> | Secret      | Secret      |
| ISA Specs      | <b>Public</b> | Often NDA   | Proprietary |

**Borg** is the only design where researchers can inspect the layout and modify logic without NDAs.

## SoC & TBDR Architecture

Borg is a complete **RISC-V SoC** featuring the **TinyQV** core. The GPU follows the Tile-Based Deferred Rendering paradigm:



Geometry is processed tile-by-tile; shaded fragments are written to an on-chip tile buffer, minimising external memory bandwidth.

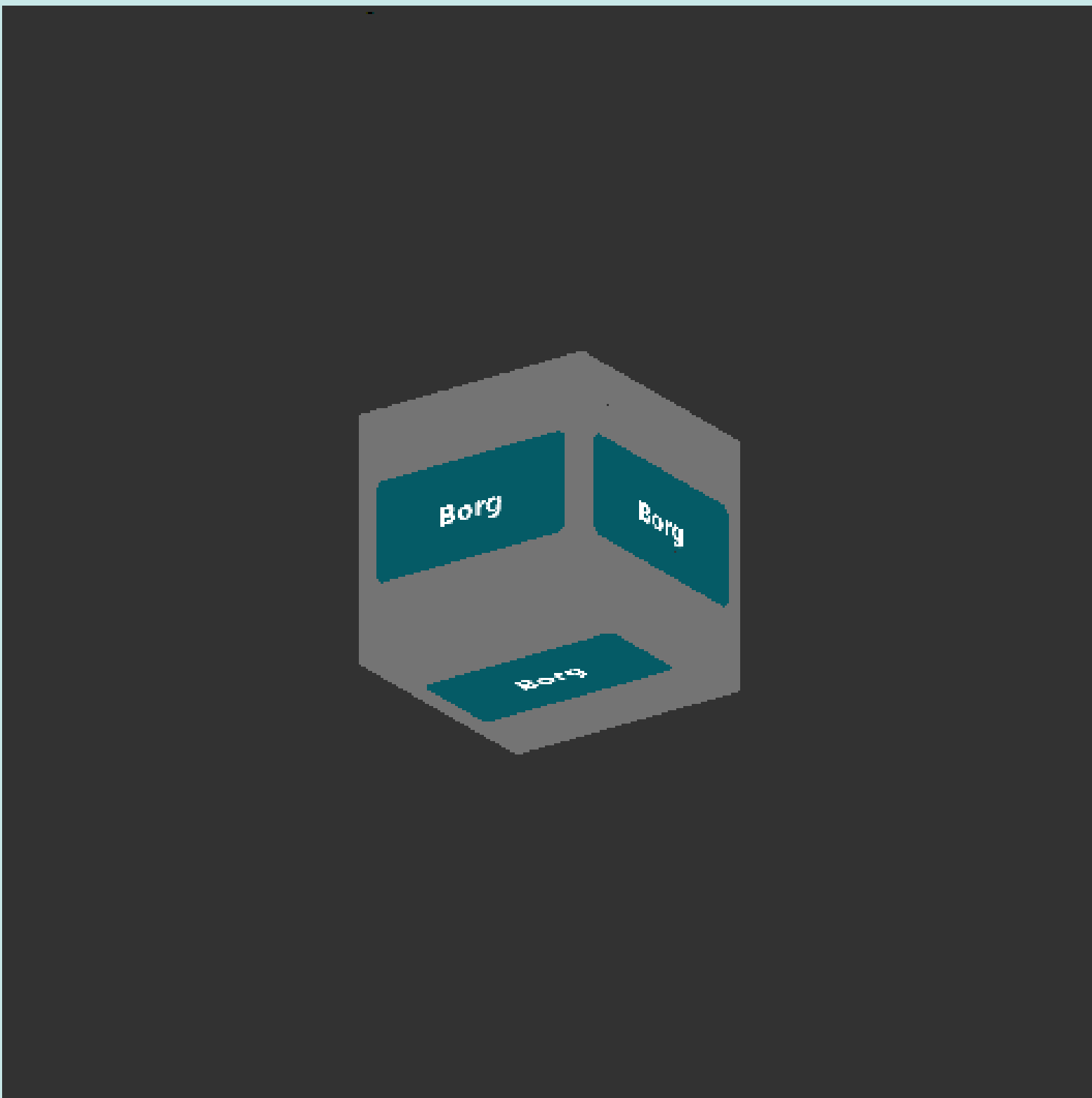
## ASIC Tapeouts (Tiny Tapeout)

| Run      | Process    | Tiles    | Content       | Cost   |
|----------|------------|----------|---------------|--------|
| TTIHP26a | IHP 130 nm | 8 tiles  | FP peripheral | 515 €  |
| TTSKY26a | SKY 130 nm | 16 tiles | Rasteriser    | 1235 € |

Chips for TTIHP26a expected July 2026.

## Current Demo

- Running on **pico-ice** (iCE40 UP5K, 5280 LCs):
- 12 textured triangles, 32×32 framebuffer
  - Full pipeline: vertex shader → texture fetch → Z-buffer → output
  - Orchestrated by integrated **TinyQV** RISC-V core



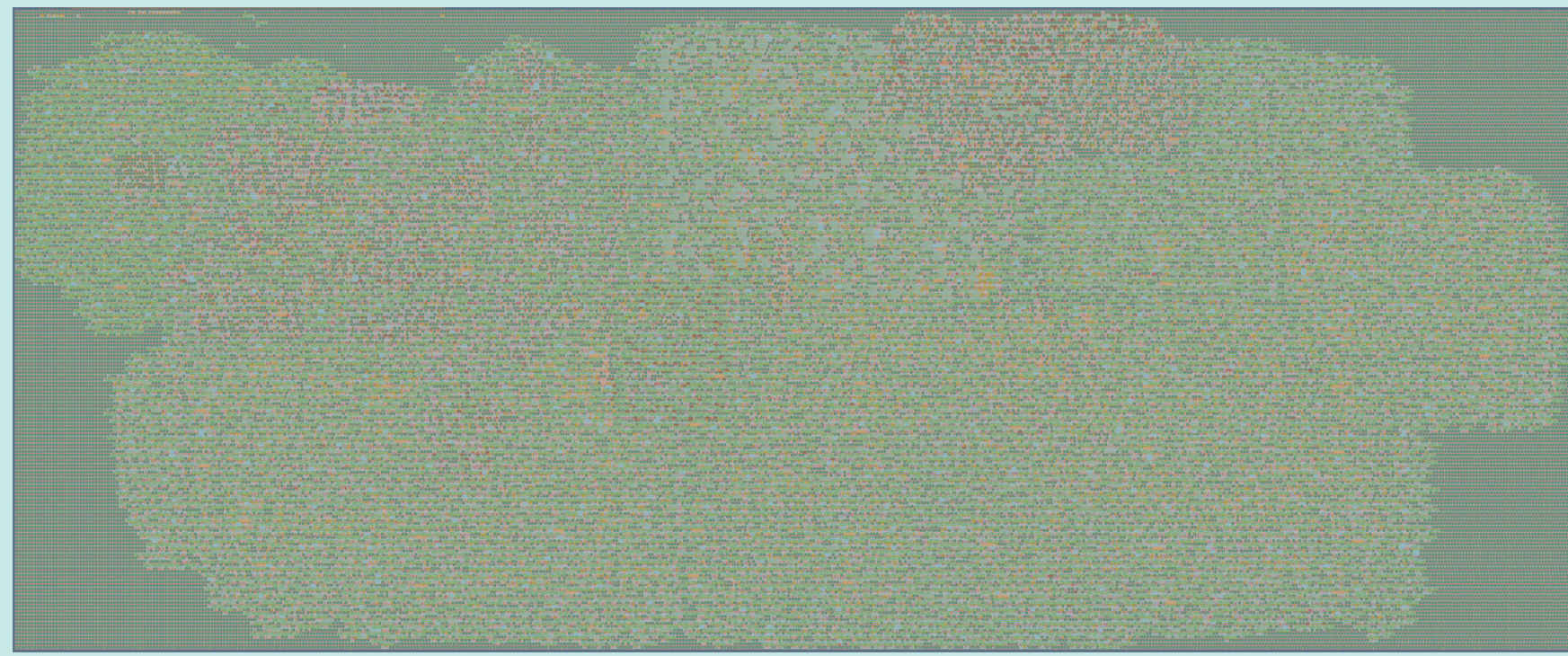
VkCube running in Verilator simulation (500×500)

**Bandwidth note:** Current bottleneck is shared PSRAM access. GPU DMA engine (Phase 3) will eliminate CPU involvement.

## Chisel: Hardware as Code

- High Productivity:** 28-day sprint made possible by Scala’s expressiveness compared to traditional Verilog/VHDL.
  - Type Safety:** FIRRTL compiler catches wiring errors early.
- “Borg proves that a single researcher can build a complete GPU pipeline in weeks.”

## Physical Design (GDSII)



| Metric          | Value (TTSKY26a)        |
|-----------------|-------------------------|
| Logic Area      | 1.4 × 1.4 mm (16 tiles) |
| Est. Gate Count | ~180k (excluding BRAM)  |
| Clock Target    | 50 MHz (Standard Cell)  |

## FPGA Resource Budget (iCE40 UP5K)

| Resource     | Used   | Available |
|--------------|--------|-----------|
| Logic Cells  | ~5 280 | 5 280     |
| BRAMs (4 kb) | ~14    | 30        |
| DSPs         | 1      | 8         |

## Open-Source Toolchain

**HDL** Chisel (Scala, **1513 lines**)  
**Synthesis** Yosys  
**P&R** nextpnr-ice40 / **LibreLane** (Nix)  
**Simulation** Verilator, cocotb, Arcilator  
**Compiler** SPIR-V → SPIR-B (**1600 lines** Python)  
**Firmware** 32-bit RISC-V (**2000 lines** C)  
**Reg. map** PeakRDL-chisel  
**Math Cores** **Berkeley Hardfloat** (FMA)

**Reproducible Build:** direnv + Nix flakes.  
Zero-install setup for all 18+ EDA tools.

**Ecosystem Impact:** Pioneers the first fully reproducible ASIC flow in **nixpkgs** (LibreLane merged Jan 2026).

## Future Work & Open Questions

- [wip] **GPU autonomy** — DMA, vertex sequencer, VGA
- ULX3S port for higher resolution rendering
- BVH ray traversal ISA extensions (path tracing)
- GONZALES path tracer integration

### Open questions — let’s discuss!

- Can a TBDR pipeline scale to multi-core tiles?
- Minimal ISA for hardware BVH traversal?
- How to best expose Borg via Mesa Vulkan?

## Funding



NGI ZERO

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## Selected References

- M. Bell, *TinyQV: A RISC-V Core for Tiny Tapeout*, GitHub, 2024.
- J. Bush, *Nyuzi Parallel Processor*, GPGPU-7, 2014.
- A. Taneja et al., *Vortex: Open Source RISC-V GPGPU*, 2021.
- Woop et al., *RPV: A Programmable Ray Unit*, SIGGRAPH, 2005.
- Bachrach et al., *Chisel: Constructing HW in a Scala Embedded DSL*, 2012.
- C. Wolf, *Yosys Open SYnthesis Suite*, 2012.
- M. Venn, *Tiny Tapeout: Open Access Silicon*, tinytapeout.com, 2023.
- SkyWater/IHP, *Open Source 130 nm PDKs*, 2020–2024.
- Berkeley HW Group, *Hardfloat: Chisel FPU Cores*, 2018.
- A. Wendleder, *LibreLane init in nixpkgs*, PR #471712, 2026.

## Comparison with Related Open GPUs

|                | Borg   | Nyuzi   | Vortex       |
|----------------|--------|---------|--------------|
| Architecture   | TBDR   | GPGPU   | RISC-V GPGPU |
| ISA            | Custom | Custom  | RV32         |
| Tapeout        | ✓      | —       | —            |
| Open toolchain | ✓      | partial | partial      |
| Target         | iCE40  | Arria 5 | Xilinx       |

## Learn More



Source Code



Book / Docs

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